Cersidy
I found the ensemble method is more accurate of the prediction as compared to linear models or a single decision tree. I figured having the contribution of multiple decision trees makes the final answer more towards the correct score compared to just a single tree or a linear model. Ensemble model strength comes from the wisdom of crowd as it takes the average answers of multiple trees.

Samuel Camacho
Reflection In this lab, I compared a single Decision Tree to a Random Forest using the Iris dataset. Both models got 100% accuracy because the dataset is small and clean, even though a Random Forest is usually better at preventing overfitting by combining votes from 100 trees. The feature importance chart also made it easy to see that petal width and petal length were the main feature

Ebuka
This lab helped me understand how ensemble methods combine multiple models to make predictions. I learned how Random Forest uses several decision trees and combines their votes to reduce reliance on a single tree. The lab also introduced the difference between bagging and boosting and explained how feature importance can help identify which measurements influence predictions. Overall, reviewing the lab helped me better understand why comparing models is important instead of assuming that a more complex model will always perform better.

Sadia Ravate

The Module 09 Ensemble Methods lab helped me understand how combining multiple machine learning models can improve predictions. Before completing this lab, I understood that machine learning models learn patterns from data, but I did not fully understand why combining models could be more effective than relying on one model. Working with the Iris dataset helped me connect ensemble learning to the practical process of training, testing, and evaluating machine learning models.

The main concept I learned was the idea of the “wisdom of the crowd.” Instead of relying on one model to make predictions, ensemble methods combine multiple models to produce a final result. I learned about bagging and boosting. Bagging trains models using different samples of the training data, while boosting trains models sequentially, with later models focusing on earlier errors. Random Forest is an example of bagging.

The lab also helped me understand the difference between a Decision Tree and a Random Forest. A Decision Tree makes predictions using a series of decision rules. A Random Forest combines many Decision Trees and uses their predictions to determine

the final classification. This can make the ensemble more robust and less likely to overfit than a single tree.

The workflow became easier to understand when I completed each stage in order. First, I imported the necessary Python libraries and loaded the Iris dataset. I then separated the data into training and testing sets using a 70/30 split. This reminded me of previous machine learning labs where data preparation and train-test splitting were important before model evaluation.

Next, I trained a Decision Tree and evaluated its accuracy. I then trained a Random Forest using 100 trees and compared its performance with the Decision Tree. Both models achieved 100% accuracy in my experiment. Initially, I expected the Random Forest to perform better because it combines multiple models. However, I learned that an ensemble does not necessarily improve accuracy when a single model already performs well on a particular test dataset.

The feature importance section was another useful learning experience. It showed me that machine learning models can provide information about which features contribute most to predictions. The visualization helped me connect numerical model output with a more understandable representation.

This lab connected to earlier machine learning concepts, including data preparation, training, testing, and evaluation. It also reinforced the importance of reproducibility through the use of a fixed random state.

One important insight was that accuracy alone does not tell the entire story. A model can perform well on one test dataset but still struggle with new data. I would like to explore additional evaluation metrics and experiment with different datasets to understand how ensemble methods perform under more challenging conditions.

Overall, this lab helped me develop a clearer understanding of how ensemble methods work and why comparing multiple models is an important part of machine learning.